Automating device assignments in Apple Business/School Manager

Using the AXM API



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From Manual to Automated

- *Before the Ability to Automate*
- *How the API Changed the Game*
- *Automating Device Assignments at Scale*



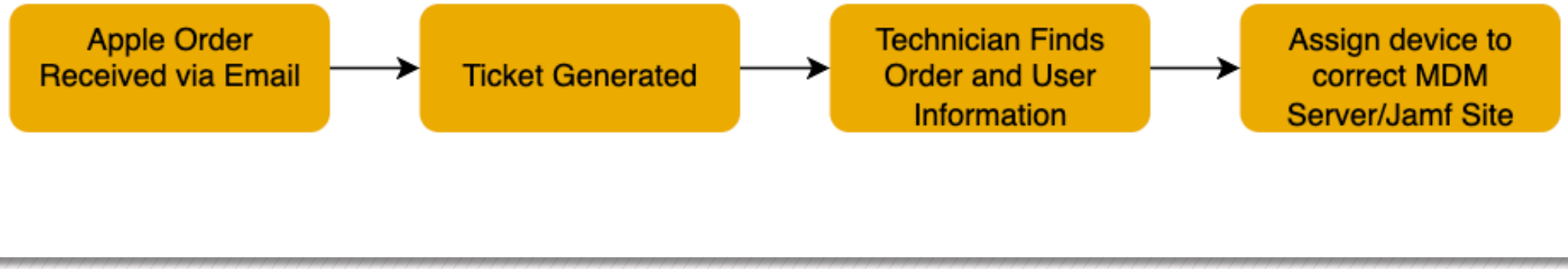
Device Assignment at Scale is Hard

- *Decentralized IT Support*
- *Multiple Jamf Pro Sites*
- *Multiple ASM Server/Locations*
- *A lot of devices…*

A lot of moving parts

- *17+ Jamf Pro Sites/ASM Servers*
- *Hundreds of orders a Year*
- *Thousands of Users*

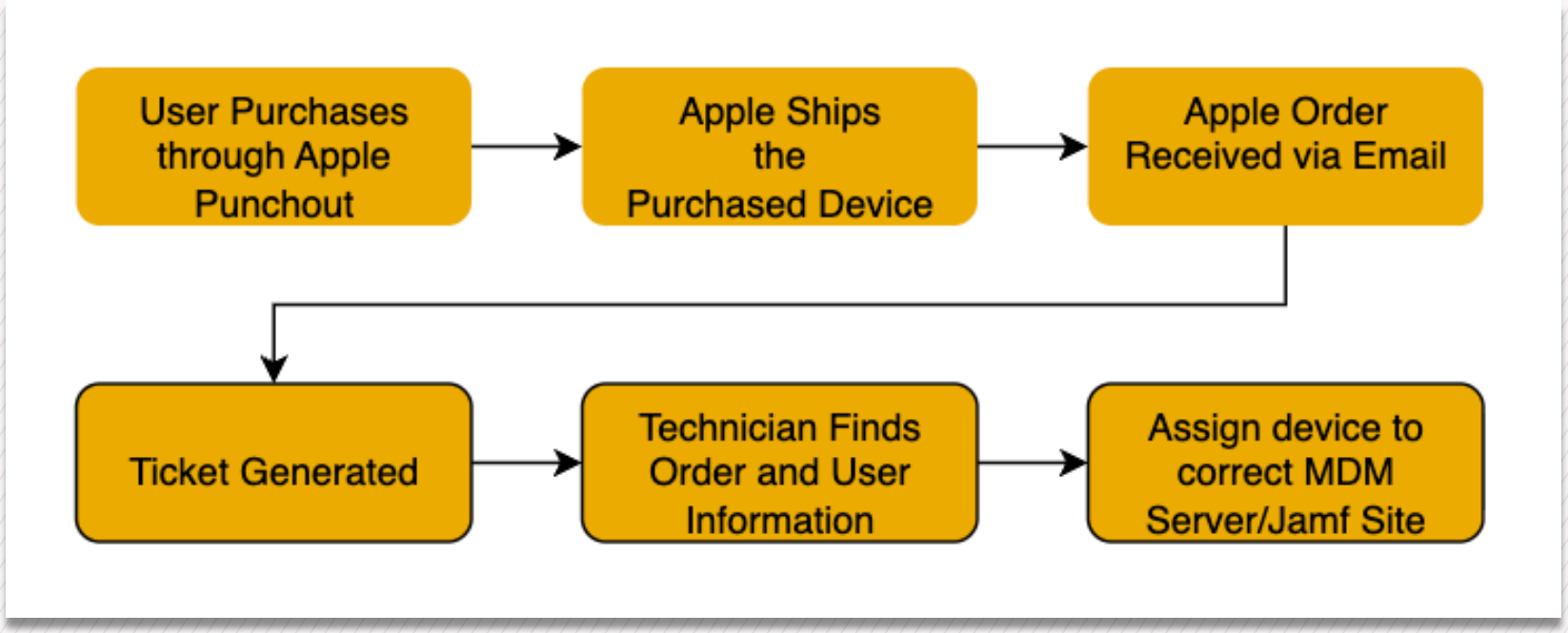
How Device Assignments looked Before the API



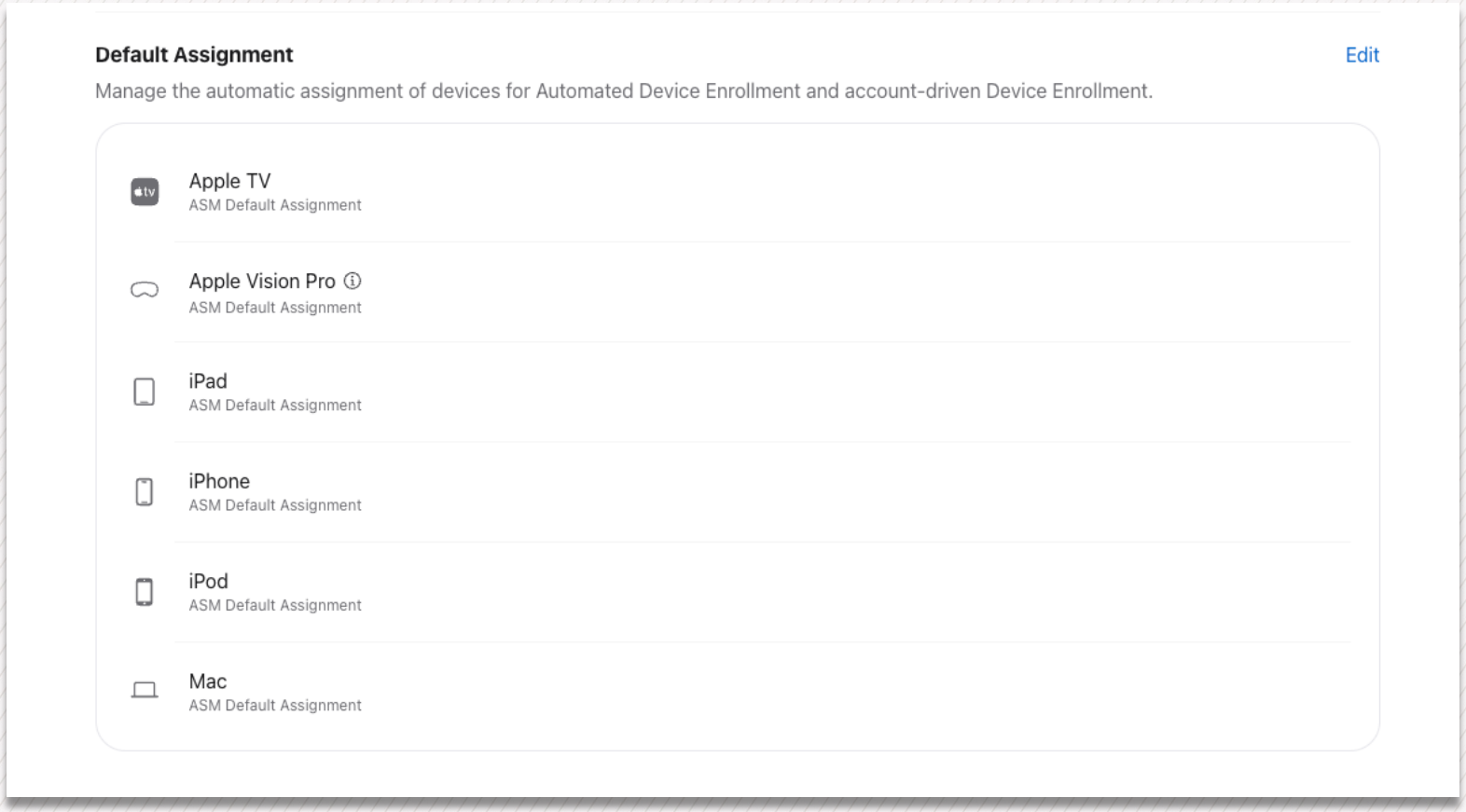
Initial Data Collection and Alerting



New Devices and Assignments



Default Assignment Location

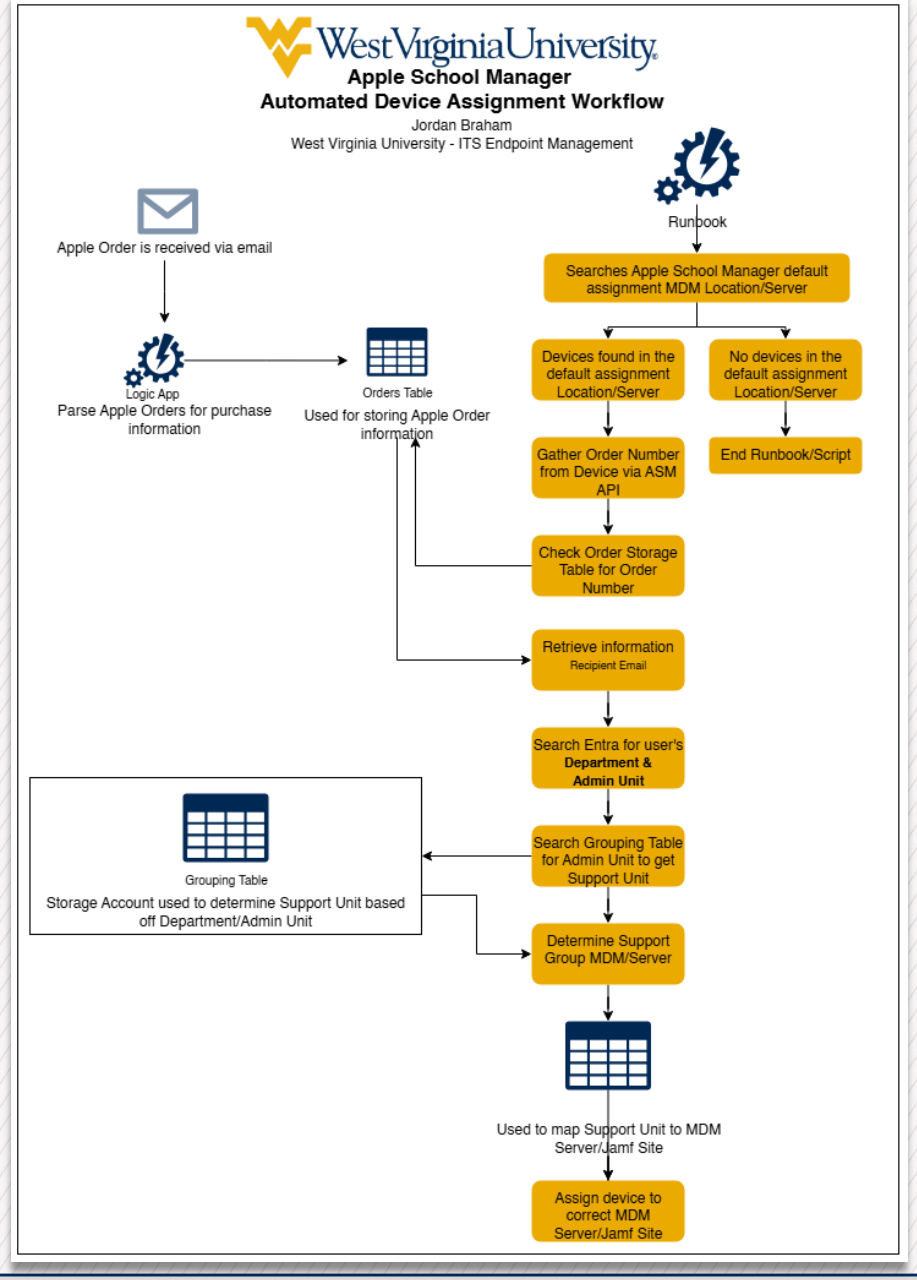


The Apple AxM API Changed the Game

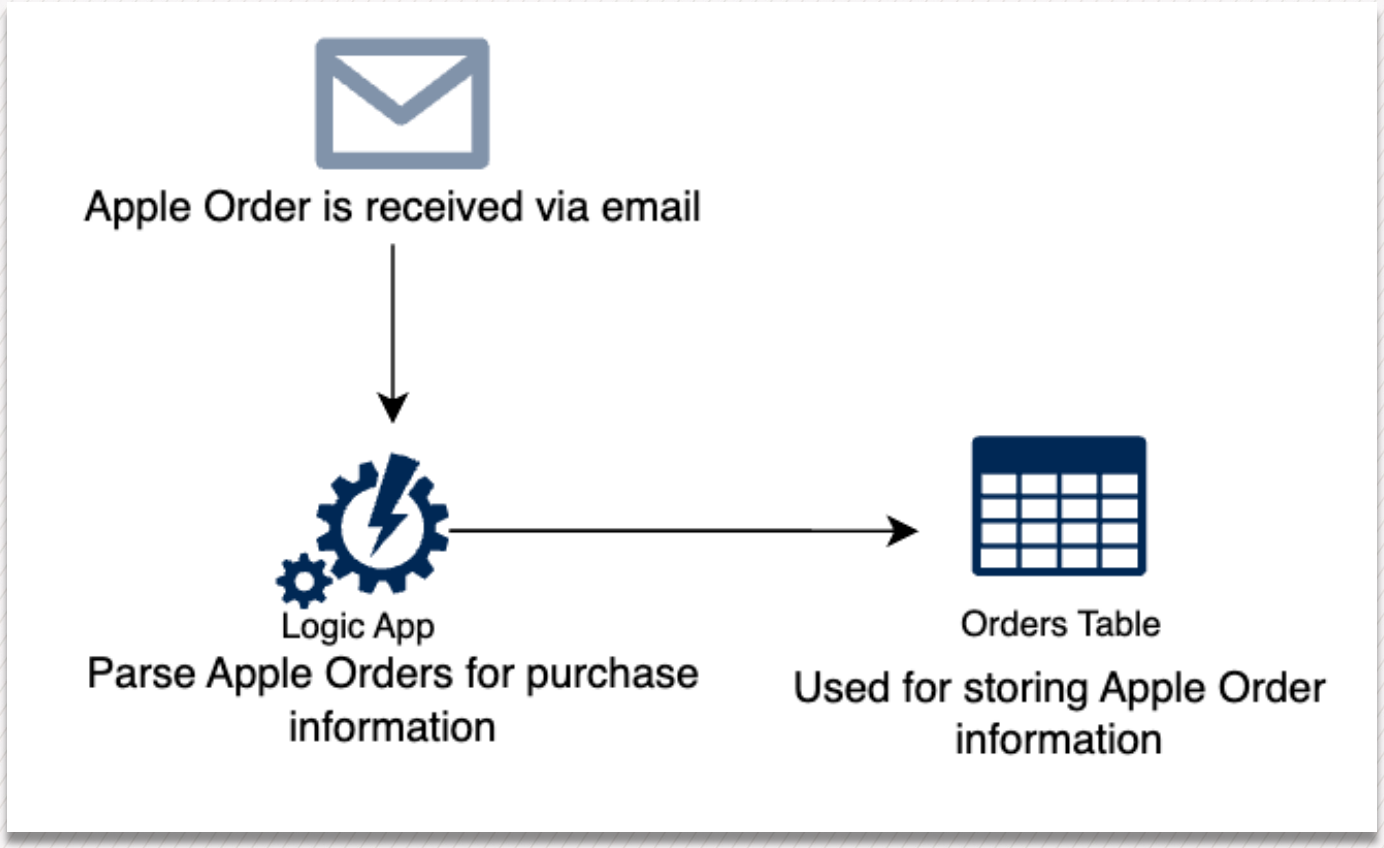
- *The ability to find devices in your AXM Instance*
- *The ability move devices between MDM Servers*

Automation of device assignments

- Cloud Hosted
- Daily runs
- Reliability

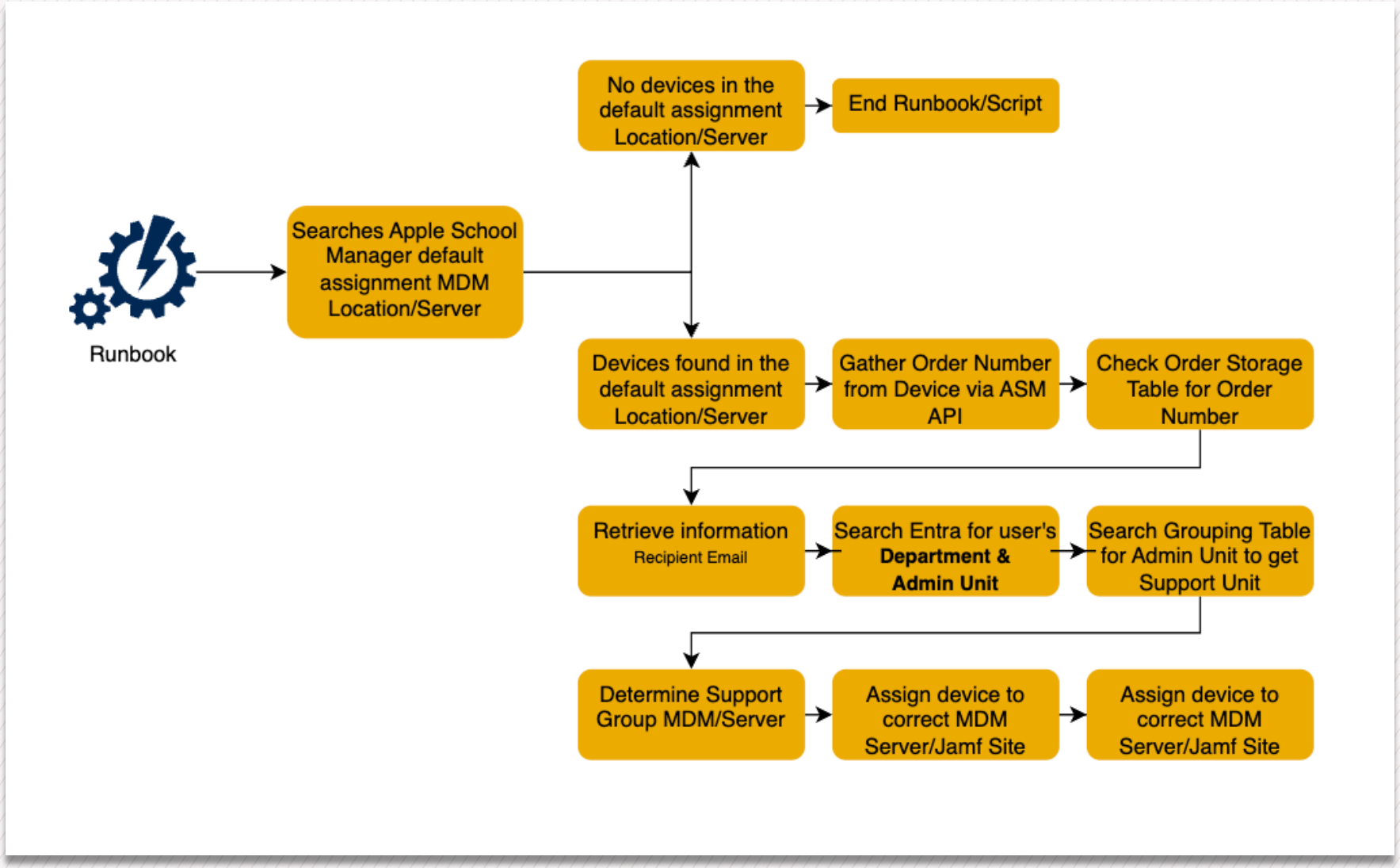


Purchase Order information




```
{
  "PartitionKey": "ASM",
  "RowKey": "AAA1234567",
  "EmailRecipient": "Foo.Bar@mail.wvu.edu",
  "PONumber": "123456789",
  "WebOrderNumber": "123456789",
  "AppleDeliveryNumber": "123456789",
  "AppleCustomerNumber": "123456789",
  "CarrierName": "DHL EXPRESS",
  "CarrierTrackingNumber": "00000000000000000"
}
```

What happens when the script runs?



Processing Devices for Assignment

URL

Default Server Assignment ID

```
GET https://api-school.apple.com/v1/mdmServers/{id}/relationships/devices
```

```
INFO: 2026-02-06 08:39:00,989: Getting devices assigned to ASM Default Assignments
INFO: 2026-02-06 08:39:00,989: Getting devices for: [redacted]
INFO: 2026-02-06 08:39:01,427: Following devices are in the ASM Default Assignment and will need to be assigned:
INFO: 2026-02-06 08:39:01,428: ['[redacted]']
```

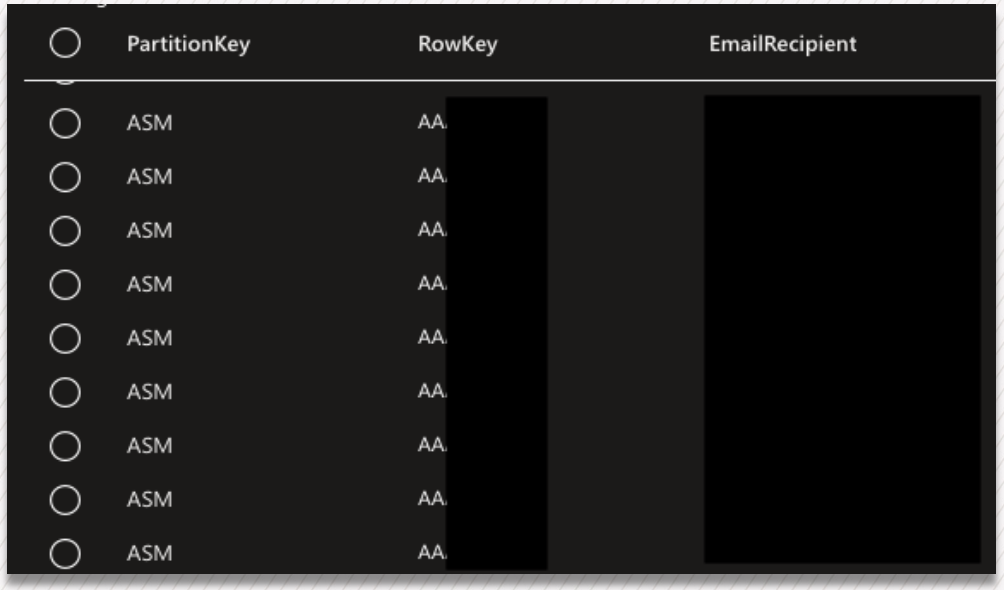
URL

```
GET https://api-school.apple.com/v1/orgDevices/{id}
```

```
{
  "data": {
    "type": "orgDevices",
    "id": "SERIAL",
    "attributes": {
      "orderNumber": "1234567890"
    }
  }
}
```

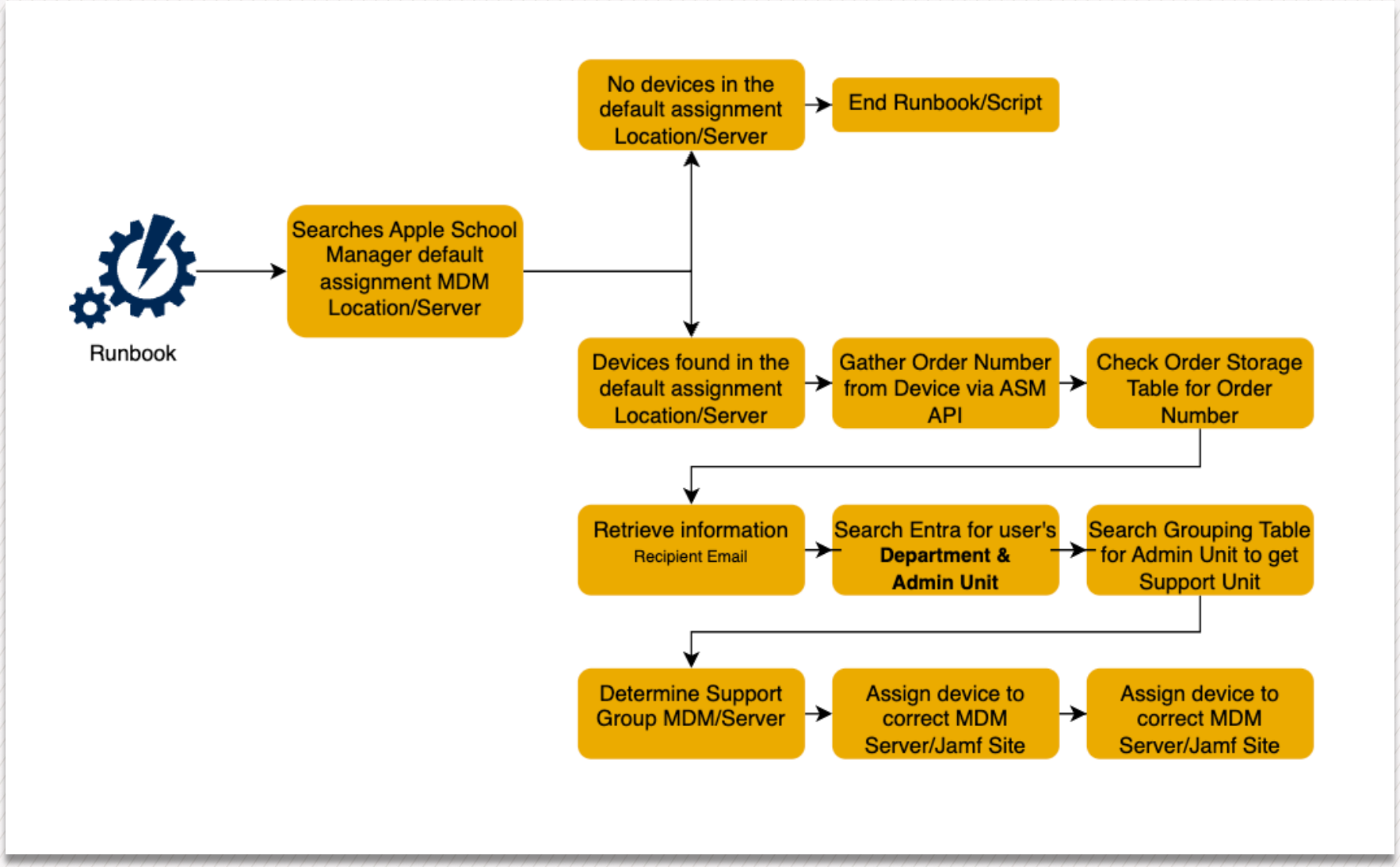
Mapping Orders to Users


```
{
  "PartitionKey": "ASM",
  "RowKey": "AAA1234567",
  "EmailRecipient": "Foo.Bar@mail.wvu.edu",
  "PONumber": "123456789",
  "WebOrderNumber": "123456789",
  "AppleDeliveryNumber": "123456789",
  "AppleCustomerNumber": "123456789",
  "CarrierName": "DHL EXPRESS",
  "CarrierTrackingNumber": "00000000000000000"
}
```

PartitionKey	RowKey	EmailRecipient
ASM	AA.	
ASM	AA.	
ASM	AA.	
ASM	AA.	
ASM	AA.	
ASM	AA.	
ASM	AA.	
ASM	AA.	
ASM	AA.	

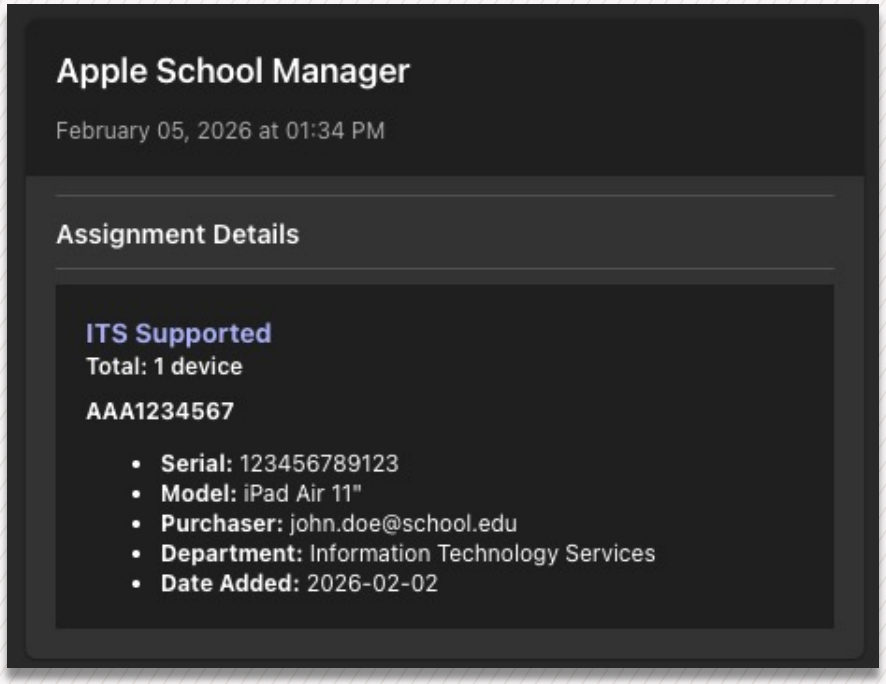
Processing Devices for Assignment



Mapping of Support Units and Jamf Sites

PartitionKey	RowKey	Timestamp	FriendlyName	ServerId
ASM	Testing	2025-06-30T18:44:48.835218Z	Testing Unassigned	00000000000000000000000000000000
ASM	Applied Human Sciences	2025-08-13T18:00:50.372257Z	College of Applied Human Sciences	[redacted]
ASM	Arts and Sciences	2025-08-13T18:01:08.1741588Z	Eberly College of Arts and Sciences	
ASM	Athletics	2025-08-13T18:01:31.2481591Z	WVU Athletics	
ASM	Business and Economics	2025-08-13T18:01:47.6039928Z	College of Business and Economics	
ASM	Creative Arts and Media	2025-08-13T18:02:02.6616229Z	College of Creative Arts and Media	
ASM	Engineering and Mineral Resources	2025-08-13T18:02:18.0120778Z	Statler College of Engineering and Mineral Resources	
ASM	Extension	2025-08-13T18:02:37.4762273Z	WVU Extension Service	

Getting Notified of Device Assignments



Notification of Device Assignments

How has the workflow impacted West Virginia University?

- *In 6 months, 200+ Orders have been processed*
- *Approximately 350 Devices Assigned*
- *99% Accuracy*
- *Over 15+ hours of manually assigning devices saved*

Thank You

- *LinkedIn: Jordan-Braham*
- *Slack: Jbraham*